

Programming Language Paradigms

- **Imperative languages**

Big ideas:

Sequence of commands.

Variables are set to initial values.

Variables' values are changed by the program as it executes.

Example languages:

FORTRAN, COBOL, C, C++, Java, Ada

Where these languages excel:

General-purpose programming

- **Functional languages**

Big ideas:

Functions and function composition

Example languages:

Common LISP, Scheme, ML, Haskell

Where these languages excel:

Artificial intelligence

- **Logic programming languages**

Big ideas:

Symbolic logic

Example languages:

Prolog, Gödel, Mercury

Where these languages excel:

Artificial intelligence

Finding all solutions that meet a set of constraints

- **Object-oriented languages**

Big ideas:

Classes and objects, implementing abstract data types

Association of methods with classes and objects

Information hiding

Inheritance

Examples of pure or nearly-pure OO languages:

Smalltalk, Eiffel, Ruby, Java

Examples of hybrid OO languages:

C++, Ada

Where these languages excel:

Large programs, especially ones that manage complex data.

- **Scripting languages**

Scripting languages frequently include these features:

- Dynamic typing
- Dynamic arrays
- Associative arrays
- Implicit variable declarations
- String and pattern operators
- Ability to call UNIX utilities

Example languages:

sh, awk, Perl, JavaScript, PHP, Python, Ruby, Lua

Where these languages excel:

- Communication between UNIX processes
- Communication between the Web and databases or programs
- Text file processing
- File format conversion

- **Concurrent languages**

Big ideas:

Features to create concurrent processes, to enable the processes to access resources safely, and to enable the processes to communicate.

Example languages:

Erlang, Clojure, Go

Where these languages excel:

Concurrent programming

- **Multiparadigm languages**

Many languages contain elements of multiple paradigms.

For example:

- * Hybrid OO languages like C++ and Ada can be used for either procedural programming or OO programming.
- * Clojure is a functional language designed for concurrent programming.
- * Ruby is an OO scripting language.
- * λ Prolog includes both functional and logic programming ideas.
- * Ada and Java contain features for managing concurrent processes, even though concurrent programming is not their primary focus.